

Designing a collaborative virtual reality system to assess social inclusion among neurodiverse students

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ABSTRACT

The following article presents a collaborative VR system designed to assess social inclusion among neurodiverse students. Students can join virtual environments to communicate and collaborate in a virtual supermarket shopping trip. A virtual classroom is developed in which students can be introduced to their task and the basic concepts of shopping. An overview of related literature in inclusion research is presented and used to discuss how the VR system can be used in future studies to measure social inclusion.

Index Terms: Virtual Reality—Social Inclusion—Interaction Design—Neurodiversity; Sociometry

1 INTRODUCTION

Experiencing social inclusion is an essential determinant to an individuals' health and well-being [1]. Social inclusion can be defined as the interaction between interpersonal relationships and community participation [2]. Social inclusion as a goal for individuals with disabilities has been associated with decreased stigma and discrimination, and improved happiness, self-esteem, confidence, and mental health [2]. However, according to statistics from the European Union, neurodivergent individuals are still at a higher risk of experiencing social exclusion [3]. It is estimated to be experienced by 28,7% of neurodivergent individuals, for whom building meaningful relationships become a hindrance to their prosperity. Stigma, discrimination, lack of access to essential services remain an issue resulting in social exclusion of neurodivergent individuals through history [4]. Attendance in elementary school activities is essential for social skill development among neurodiverse students. We describe neurodiversity in a group of individuals (e.g., in a school class or later in work life) as encompassing neurotypical and neurodivergent individuals. The most common neurodivergent conditions are described as ADHD, ASD, dyslexia, Down syndrome, and cerebral palsy [5]. A neurodivergent student provided with all special education services within the general education classroom can result in full inclusion, whereas a student who sometimes has to be removed from the classroom may experience partial inclusion [6]. By allowing a student from an otherwise self-contained special education setting to join the general education classroom, a teacher can use various strategies to promote social skill development, as well as academic skill development [6]. Educating neurotypical students on neurological differences before introducing neurodivergent students to the class and promoting awareness of preferences in interaction types can be used by teachers as strategies to increase peer acceptance [6]. Emerging technologies, such as VR, have potential to address the challenges faced by neurodiverse users [5]. Motti propose that these technologies are promising in assisting neurodiverse users with executive functions, communication and social skill development [5]. Currently, several social platforms (e.g., VRChat or Rec Room), includes rooms, avatar system, verbal and non-verbal communication,

however, not designed around DLS training or mathematics education and does not include metrics on social interactions between users

The following article describes a collaborative VR system designed to assess social inclusion among neurodiverse students. The collaborative VR system combines daily living skills (DLS) training and basic mathematical concepts in a virtual classroom and virtual supermarket. The approach of transitioning from the classroom to a concrete daily situation, such as shopping, may deem to possess qualities regarding knowledge transfer relevant to social and academic skill development. Knowledge transfer is a term commonly used to describe the learning outcome of a digital application in which students learn in a virtual environment (VE). In this sense, it becomes important for researchers to evaluate if knowledge is understood to a degree in which the individual can apply it in real life.

2 RELATED WORK

Digital technology, such as digital applications or services, can be used as a medium and tool for teachers to support and facilitate social activities. This is allowed by the delivery of real-time services that can enable individuals to learn, work, travel, socialize, shop, and interact with the community without being subject to physical barriers [7]. Reducing social gaps between neurodiverse individuals by encouraging and supporting social inclusion is enabled by digital technology, which improves individual's quality of life [7]. Moreover, Motti & Evmenova (2020) [8] state that neurodiverse individuals could largely benefit from innovative technologies and the customization of these to the individual needs of neurodiverse users.

The increasing accessibility of high fidelity VR technology allows the creation of VR-based tools for social and academic skill development. Engaging multisensory stimuli, embodied interactions and communication allow users to experience social settings in entirely new ways with great possibilities in customization of user preferences. While the sensation of presence may become a means to increased enjoyment of educational activities [9], the social interaction and communication may reduce stigma and discrimination, due to the awareness of the virtuality of the situation and by perceiving users through their custom avatars.

Wasserman et al. [10] explored how to design inclusive VR experiences for neurodiverse students to enable inclusive play. The study describes the experience of three families with neurodiverse children that used a VR application consisting of 3 VEs. The Mechanics-Dynamics-Aesthetics (MDA) framework [11] was used to map the effects of passive and active interaction styles on the perceived outcome (i.e., aesthetics) of the VR experience. The article concludes that the MDA framework can help developers build experience-driven games [10] by evaluating the aesthetic experience of participants through a VR application's implemented mechanics. A similar approach to designing collaborative experiences was presented in [12], with the focus on understanding dynamics of asymmetric VR.

Boyd et al. [13] explored the support of variations in sensory processing for diverse sensory-perceptual abilities found in individuals with ASD. The study describes this as an opportunity to improve the

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lives of individuals with ASD through their interaction with technology and their surrounding world. In regard to communication and social skills development in VR, the authors describe how VR environments can be advantageous for individuals with ASD, as unique situations can arise, that normally do not happen in strict educational settings [13]. This can be beneficial as a process of acclimatization to unplanned situations that all individuals are faced with in the world outside of school.

Bob's Fish Shop is a VR system designed to help children with ASD practice typical social interactions and conversational skills [14]. The VR system situates the user in a fish shop designed for training of verbal communication with a scripted non-player character (NPC) and non-verbal communication by responding to gestures of the NPC such as waving. The VR system estimated user's joint attention from gaze tracking data and rule-based artificial intelligence [14]. An important quality of the VR system is its ability to practice social skills in a safe and controlled environment [14]. For some students, this may still be a preferable experience for slowly introducing activities for social skill development. One could imagine that in time, a neurodivergent student may use this type of VR system before being introduced to the social VR system we present, which can be seen as similarly safe, however, with much less control over events, as they are not scripted and occur dynamically according to the actions of users.

Moustafa & Steed [15] conducted an exploratory social VR study, which showed a strong propensity for users to feel present and engaged with group members. The study points out that users experience a range of emotional states in VR, similar to face-to-face experiences. Moreover, relevant for modifying behaviour related to social inclusion, the study also found that existing group dynamics are transferred to VR. This finding is interesting to inclusion research in VR, as the goal commonly is to change group dynamics (e.g., in a neurodiverse school class). A similar study by Riva et al. [16] investigates VR as an effective medium. The article's analysis confirms the efficacy of the use-case with examples of an "emotional" environment eliciting a similar emotional response in participants. Another study by Pan et al. [17] highlights the prospects of using VR to study social interaction between humans. The article describes experimental control, reproducibility, and ecological validity as major advantages of VR in psychology research. Moreover, the article describes current challenges related to embodiment, uncanny valley, simulator sickness, presence, ethics, and experimental design. Roth et al. [18] investigated avatar realism and social interaction quality in VR. The study found that non-realistic avatars (e.g., a mannequin) tend to impede social interactions.

Brooks et al. [19] used a VE of a kitchen to train individuals with learning disabilities. Twenty-four students participated in an experiment, which showed that virtual training can be as beneficial as real training and more beneficial than using a workbook and no training. Adjorlu & Serafin have conducted studies showing the potentials of using VR to teach shopping [20] and money management skills [21] to students diagnosed with ASD. A review on VR rehabilitation of individuals with intellectual disabilities found that existing literature has primarily been aimed at promoting skills for independent living, enhancing cognitive performance, and improving social skills (REF). The article emphasizes the importance of considering and addressing knowledge transfer as an outcome of VR rehabilitation [22]. Based on the ideas and findings presented in this section, the following sections will describe and discuss the design of the collaborative VR system for assessing social inclusion among neurodiverse students. The VR system includes verbal and non-verbal communication in a combined DLS and math training simulation of a virtual supermarket. Users can join and collaborate on performing the task of shopping in a supermarket. Concepts involved in performing the task can be introduced in the classroom VE of the VR system. Initially, the VR system was intended for

use on-site in schools, to evaluate outcomes in collaboration with teachers, however, the need and potential of using the VR system remotely is more present following the global pandemic of COVID-19, which in many cases results in students having to attend school remotely. The networking infrastructure of the VR system allows for this use-case, especially due to the implemented Voice-over-IP (VoIP) communication protocol.

3 DESIGN

We combine DLS training in a virtual supermarket with social mechanics to create a collaborative experience between neurodiverse students. We followed guidelines from the research community [23] in determining the needs of the educational material in the VR system for 6th grade lower secondary school students. For the neurodivergent student, practicing and mastering DLS is an important factor to well-being. In accordance with the Danish KOM-framework on mathematics education, the mathematical concepts of the equal sign and unknown variables is introduced at 6th grade. Thereby, we intended to combine DLS and math training in the experience, to challenge both neurotypical and neurodivergent students. Shopping in a supermarket is an essential skill to possess in order to live independently. Conceptually, shopping in a supermarket can involve several basic mathematical concepts, such as magnitude, multitude, arithmetic, and algebra. Each student is represented in VR by a virtual avatar selected by themselves. The virtual avatar's body uses inverse kinematics to respond appropriately to the head and arm movement of the user. A student can verbally communicate with participating peers by using the embedded microphone. The transmitted audio of the speech is spatialised to provide realistic sound behaviour in the VEs, but also to limit distractions from the overlapping conversations that may occur between users. From the mapped motions of the virtual avatar, users can express themselves non-verbally through body language and gestures. The following sections will describe the design of the VEs.

3.1 Virtual environments

The VR system allows students to enter three different VEs: avatar selection (see fig: 2), classroom (see fig: 1), and supermarket (see fig: 4). The flow of the VEs was designed to allow students to choose a virtual avatar, receive an introduction to the concepts, and initiate training. A teacher has additional system control to transition students collectively between classroom and supermarket in case this is needed.

3.1.1 Avatar selection

The first VE includes a virtual mirror and a spatial UI for avatar selection. The student can choose from ten premade avatars varying in bodily features and clothing (see fig: 2). Additionally, the UI for choosing an avatar also includes a height re-calibration button to ensure that the virtual avatar is aligned with the height of the student. After picking an avatar and adjusting height, the student can press a spatial UI button to transition to the classroom VE.

3.1.2 Classroom

The classroom incorporates interactive objects and concepts from the supermarket (see fig: 1). Here, the teacher can introduce students to the interaction and concepts of the supermarket. The concepts revolve around the virtual balance, which can be used to teach the basic idea of the equal sign and variables in an equation.

The balance deflects based on the comparison of weight sum from objects placed on each balance pan. A number of products, such as fruits and vegetables, can be used to set up an equation on the balance. Products are grabbed and placed on balance pans using the VR controller. Moreover, a task pad with predefined questions can be used as inspiration for the teacher. A pointer from the VR



Figure 1: An overview of the virtual classroom.



Figure 2: The virtual mirror and the 10 avatars.

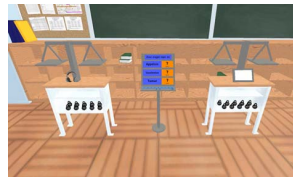


Figure 3: The virtual balance and task pad.



Figure 4: An overview of the virtual supermarket.



Figure 5: The basket and shopping list as seen in the virtual supermarket.



Figure 6: The virtual balance of the supermarket.

controller is used to select questions to note answers using a virtual keyboard (see fig: 3).

3.1.3 Supermarket

The supermarket VE includes four steps in shopping products: identify, collect, scan, and pay. Students are given a shopping list at entrance of the supermarket.

Each item can be found in the supermarket and either requires counting or weighing to collect the correct multitude or magnitude of products. The students are met with a particular collaborative task in the supermarket; fruits and vegetables have to be weighed to find the right amount. A virtual balance can be used to fulfill this task (see fig: 6). Users can communicate about potential solutions before approaching the virtual balance.

The virtual balance is accompanied by small weights of incremental weight by the hundreds. After reaching a solution, students can pack the fruits or vegetables in a plastic bag. At the self-scanning register, students can scan items, remove items, or clear the list and finally, pay for the purchase. A wallet with coins and notes can be found by the counter and can be placed in the register to accumulate the correct payment amount.

If the task is completed, a rewarding confetti visual effects (VFX) animation with a short musical melody was played, and if the task was partly completed (i.e., not all items from the shopping list acquired), a bell-like sound was played.

3.1.4 Interaction with the VEs

Students can interact with the VE in different ways. There are two basic ways to interact with the VE: grabbing and pointing. Grabbing is primarily used in solving tasks, as they require users to move and collect interactive objects. A student can also interact with UI elements (e.g., in avatar selection or classroom activities) by enabling a pointer from the controller.

Teachers can join the VEs together with their students, enabling them to help and tutor their students in the tasks of the VEs. However, asking for help may be problematic for some students. Moreover, there may exist different preferences among neurodiverse students in the presentation of help. Therefore, each avatar was equipped with

a wristwatch containing visual and auditory help. Visual help can be more descriptive, in terms of task execution, but requires more attention, creating difficulties in multitasking while learning about and performing the task. At first, this help-module is used to provide additional discrete support for students in the learning activities. In future iterations, this feature can also be used to track student's preferences in learning activities, which is useful knowledge for the teacher.

4 MEASURING THE OUTCOME

Measuring the outcome of a VR intervention for assessing social inclusion requires evaluation of several factors of the experience.

Firstly, usability has partly been addressed in a separate exploratory evaluation on a previous iteration focusing on collaborating in the virtual supermarket. Seven neurodivergent users and one teacher participated in an exploratory study on the use of the VR system. Several usability issues were identified. For example, some students experienced an issue with the plastic bag, which unfortunately caused confusion during this phase of the experience. However, with assistance from the teacher, the students were able to continue and fulfill the task. Moreover, the lack of distinctive visual features of avatars were commented on by test participants. Customization of avatars was found to be important from the feedback of the users, since they otherwise had a hard time distinguishing users purely from auditory cues.

Finally, the test also showed the importance of considering roles in the collaborative DLS training task. Asymmetric mechanics (e.g., a difference in unique user abilities) of the VR system should not be enforced upon users, as that may cause undesired exclusion of users that dislike their assigned set of mechanics, however, the environment should implicitly invite to role-taking on a more voluntary basis. The roles of the supermarket are not pre-defined, however, certain mechanics such as the basket, shopping list, locomotion and interaction techniques, and the virtual balance seemed to have been accepted by the neurodivergent students.

Following the identification and remediation of usability issues, the system's training abilities should be assessed. This should be

compared with literature on user studies suggesting that DLS training in VR is beneficial. The methodology here would consist of measuring a neurodivergent student's ability to shop in a real supermarket following the virtual intervention [24]. Neurotypical students' mathematical understanding and competencies following the virtual intervention can be assessed from tracing knowledge transferability. In this sense, characteristics of the VR system can be traced by the teacher when a student solves similar tasks in classroom activities. In previous research, this has been done by observing a student's vocabulary while reasoning for terminology referencing the experience of the VR system [25]. Finally, the social outcome will be the primary goal of future studies on the VR intervention described in the paper. As described by Smoot (2004), previous research has established sociometry as a reliable and valid methodology to measuring the social outcome of inclusion interventions [6]. Sociometry allows researchers to quantitatively assess and map social relationships in a group. This is usually done by explicitly interviewing individuals to report preferences in their social relationships. As an example, this could be done by asking an individual about whom they would prefer to engage in a social activity with. This can be used to establish a link between two entities in a social group, which can be adjusted and refined over time. Such a "social map" can be computed and visualized through a sociogram. A sociogram is a node-based graphic representation of individuals and their social relationships.

Links may be defined by positive spatial interactions (e.g., handing over objects, gesturing at each other) and collaborative role-taking and task execution (e.g., role-taking in shopping task). Designer may link certain VR mechanics to roles, for example, the "carrier" managing basket and shopping list, the "searcher" who finds and grabs products, or the "calculator" who inspects mathematical problems related to the virtual balance or payment of products. Researchers should consider the imaginable roles of the collaborative DLS training scenario to potentially improve the user experience and increase inclusion. A mismatch between number of users and required roles of the DLS training task may create unintended social exclusion in the VE. Thereby, a social map of VE users, their role-taking, and social links can be generated to report to teachers for better understanding of the social dynamics of a group of neurodiverse students. Future research should also include the perspective of the teacher in the evaluation. Evaluation should assess the usefulness perceived by teachers compared to traditional means of explicitly interviewing students about social preferences.

For the academic skill development outcome of using the application, it will be important to keep each individual's competency level in mind. Depending on whether a neurodivergent student experiences partial or full inclusion in a school, they may have lower academic expectations. The social skill development is still valuable, even when the academic outcome is minimal [6]. However, as a means of engaging the neurotypical user, it may still be interesting to assess their academic learning outcome of using the VR intervention. One approach could be to compare the VR intervention with a traditional non-immersive learning intervention, exploring the advantages and disadvantages of VR; for example, in terms of knowledge transfer, retention, motivation. Regarding the current educational material of the VR system, it may be relevant to assess the potential of developing communication and reasoning competencies in mathematics from experiences with the collaborative VR system.

A recently published article highlights the implications of COVID-19 for researchers evaluating immersive experiences [26]. Closed research laboratories and ethical concerns regarding the reuse of equipment have become a hindrance for conducting user studies. The article proposes three solutions varying in time frame; use lab personnel and infrastructure (short-term), recruit external users who have the necessary hardware (medium-term), and generate pools of users through funded hardware distribution (long-term) [26].

Combining recruitment of users with distribution of hardware, clear instructions, and embedded behavioural data recording is a feasible plan for the current study. At this point, this data logging module involves a custom implementation by the authors. However, in the future, it is desired to use an existing framework for improved validity, such as *The Unity Experiment Framework* for studying human behaviour with VR [27]. The future works of the project will involve recruitment of neurodiverse students for the distribution of hardware and software.

5 DISCUSSION

Neurodiverse schools encompass a wide range of students with neurological differences. Because of these neurological differences, educational materials' pedagogical approaches and academic expectations also differ. VEs are highly customizable and can be tailored to students' individual needs – whether it entails reconfiguration of sensory stimuli, additional support and guidance, socialization, or privacy. Based on existing literature, we designed a collaborative VR system for neurodiverse students. The VR system presents tasks for social and academic skill development; however, the purpose of this article was to describe the potential in gaining insight into the social dynamics between users through use of the system. DLS are crucial to the well-being and independence of neurodivergent individuals. In order to keep neurodiverse students engaged, the VR system was designed to incorporate both DLS training (i.e., supermarket shopping) primarily aimed at individuals with ASD with basic mathematical concepts (i.e., equality through a virtual balance model) primarily aimed at the neurotypical individual. Together with avatar representation and verbal communication, the neurodiverse students can collaborate on executing a shopping trip for a predefined list of items. This training task can slowly be introduced in the classroom VE, which incorporates similar elements that need to be understood before executing the shopping task in the supermarket.

Moreover, we presented reflections on the behavioural observations embedded in the VR system. The VR system may serve as a provider of insight to teachers or other tutors of a class. These behavioural observations may be used to link and map relationships in the neurodiverse class.

A preliminary exploratory study showed that visual cues are important for communication among neurodivergent individuals. Auditory cues (i.e., 3D spatialisation) are not sufficient for distinguishing speakers, and therefore, accompanying visual cues in the form of mouth animation or VFX indicators are presumably needed. The test participants also commented on the need for customizable avatars to further improve their ability to distinguish speakers. In future studies, it should be investigated how we provide control and safety for neurodivergent students in dynamic social situation like the one our VR system presents. VR may be considered to elicit less harmful or traumatic experience than similar real life encounters, however, we need to study mechanisms that reduces discomfort and promotes respectful behavior in social VR. For example, this could be enabled by providing users with toggles for other players of the environment, proximity monitoring and visual communication cues.

6 CONCLUSION

The article presented a VR system for assessing social inclusion among neurodiverse students. The concept was developed from identified prospects of the existing literature, which has guided the design and development of the VR system. This resulted in the combined DLS and math training scenario, in which neurodiverse students' collaboration over time can be used to provide insight to the social dynamics of a group of students. A preliminary exploratory evaluation focused on usability and user experience of the virtual supermarket marked certain design issues related to the quality of communication and collaboration in the VE. The neurodivergent test

participants commented on the need for customizable avatars and visual communication cues to improve awareness and coordination of the tasks in the collaborative VR system. Furthermore, we discussed appropriate evaluation approaches to measuring the outcomes of the experience. DLS and academic skill development potentials were highlighted. More importantly, approaches from sociometry on the evaluation of social inclusion was presented, with reflections on the technical and practical aspects for educational designers and teachers. Finally, the future works of assessing social inclusion and plans for bypassing implications of the COVID-19 pandemic were discussed.

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